

A no-go for perturbative baryogenesis in cosmologies with a bounded thermal history

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Abstract

We show that in any cosmology whose entire thermal history satisfies $T \leq T_{\max} \lesssim 130$ GeV — bounce models with a QCD-scale density cap, low-reheating scenarios, and cyclic models — the baryon asymmetry cannot be generated by perturbative baryon-number-violating interactions of the thermal bath. For the representative dimension-six operator $(qqq\ell)/\Lambda^2$, cosmological relevance at $T_{\max} \sim \Lambda_{\text{QCD}}$ requires $\Lambda \lesssim 3 \times 10^6$ GeV even with a maximal thermal chemical-potential bias ($\epsilon_{\min} = 1.4 \times 10^{-9}$), while proton stability ($\tau_{p \rightarrow e^+ \pi^0} > 2.4 \times 10^{34}$ yr) requires $\Lambda \gtrsim 3 \times 10^{16}$ GeV: a gap of ten orders of magnitude in Λ , i.e. forty orders in τ_p . We further show that environmental “gating” of the operator cannot rescue it: power-law gates tied to the chiral condensate cannot bridge the required 10^{-22} amplitude suppression and are turned *on* inside nucleons in bag-like descriptions, while derivative (phase-velocity) gates are radiatively unstable. The surviving mechanisms are classified exhaustively: (i) B – L -conserving transfer to a dark sector with kinematic nucleon protection (asymmetric-dark-matter/mesogenesis class), (ii) condensate-carried baryon number (Affleck–Dine class), and (iii) non-thermal populations with $E \gg T_{\max}$ (heavy-relic decay, collision-induced sphalerons, black-hole evaporation). As a corollary, published bounce-baryogenesis scenarios that assume equilibrium B -violation below the sphaleron threshold are excluded unless recast in one of these three classes; in cyclic bounded- T models a primordial asymmetry is additionally excluded by entropy dilution, forcing per-cycle regeneration.

1 Introduction

2 Setup and the theorem

2.1 Bounded thermal histories

2.2 The EFT window

The lowest-dimension $\Delta B \neq 0$ operator, $\mathcal{O}_6 = (qqq\ell)/\Lambda^2$, has thermal rate $\Gamma \sim T^5/\Lambda^4$. Cosmological relevance requires $\Gamma/H \geq \epsilon$ with the efficiency floor set by the maximal thermal bias $\mu \sim \pi T$ (spontaneous baryogenesis à la Cohen–Kaplan):

$$\epsilon_{\min} = \frac{\eta_{\text{obs}}}{\eta_{\text{eq}}^{\max}} = \frac{8.6 \times 10^{-11}}{45/(12\pi g_*)} = 1.4 \times 10^{-9}, \quad \Lambda \leq \Lambda_{\text{viab}}(T_{\max}) = \left(\frac{T_{\max}^3 M_{\text{Pl}}}{1.66\sqrt{g_*}} \right)^{1/4} \epsilon_{\min}^{-1/4}. \quad (1)$$

Proton stability requires $\Lambda \geq \Lambda_p = (\tau_{\text{SK}} m_p^5)^{1/4} = 3.0 \times 10^{16}$ GeV. At $T_{\max} = 432$ MeV, $\Lambda_{\text{viab}} = 3.1 \times 10^6$ GeV: the windows are separated by 10^{10} in Λ (Fig. ??). The “EFT desert” closes only at $T \sim 10^{13}$ GeV — academically, since the SM’s own sphalerons rescue baryogenesis already above ~ 130 GeV.

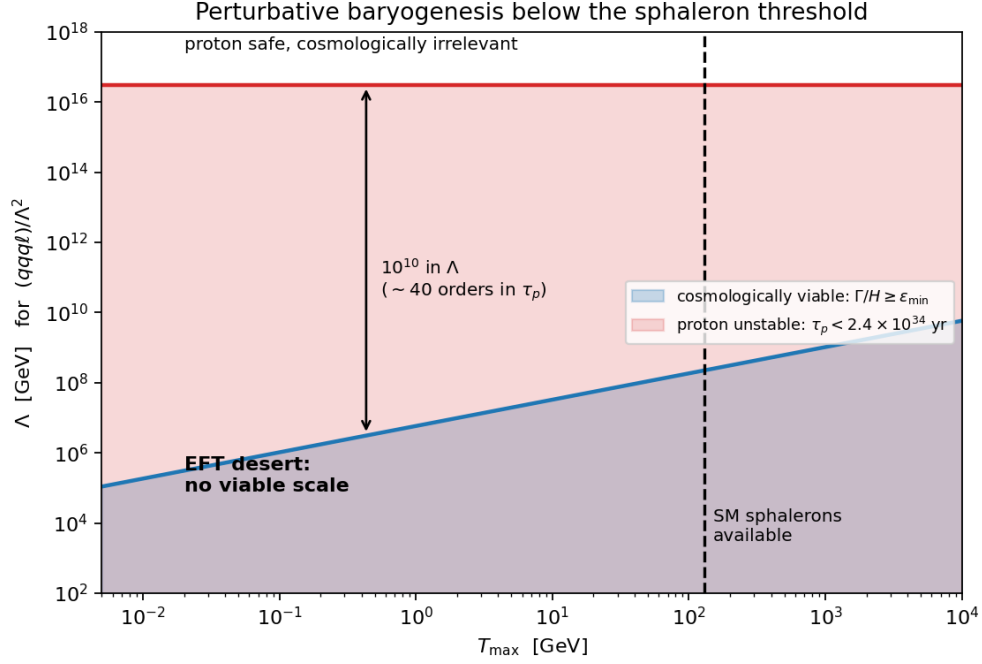


Figure 1: The $(T_{\max}, \Lambda)$ plane for $(qqq\ell)/\Lambda^2$. Blue: cosmologically capable of generating η_B under a maximal thermal bias. Red: proton lifetime below the Super-Kamiokande bound. The regions never meet below the sphaleron threshold (dashed).

3 Gating does not help

3.1 Environmental (condensate) gates

3.2 Derivative gates are radiatively unstable

4 The surviving mechanisms

4.1 Class I: $B-L$ -conserving dark transfer

4.2 Class II: condensate-carried baryon number (Affleck–Dine)

4.3 Class III: non-thermal high-energy populations

5 Corollary: cyclic bounded- T cosmologies

6 Discussion

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